BloodLink Smart Blood Donation Management System

Project Charter

Project Type :- Web-Based Information System
Project Domain :- BloodLink-Smart Blood Donation Management System
Project Status :- Completed
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Document Version :- 1.0
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1.Project Overview

The BloodLink – Smart Blood Donation Management System is a web-based application designed to simplify blood donation management. It connects blood donors, hospitals, and blood banks. The system helps manage blood donations, blood inventory, blood requests, emergency blood requests, appointments, donation camps, notifications, and blood donation-related information through an AI assistant.

2.Project Purpose

The purpose of the BloodLink system is to simplify blood donation management by connecting blood donors, hospitals, and blood banks through a single web application. It aims to make blood requests, blood inventory management, and emergency blood requests easier and more efficient.

3.Project Objectives

1. To manage blood donor information and donation records
2. To maintain blood inventory information
3. To manage blood request for hospital
4. To handle emergency blood request
5. To manage blood donation appointment
6. To provide notifications about blood donation activity
7. To provide blood donation - related information through an AI assistant

4. Project Scope

4.1 In Scope

The BloodLink system includes the following features:

1. **User Management:** Manage user registration, login, and user information.
2. **Blood Donor Management:** Manage blood donor information and donation records.
3. **Blood Bank Management:** Manage blood bank information.
4. **Blood Inventory Management:** Maintain blood stock information and blood types.
5. **Blood Request Management:** Allow hospitals to manage blood requests.
6. **Emergency Blood Requests:** Manage urgent blood requests.
7. **Appointment Management:** Manage blood donation appointments.
8. **Donation Camp Management:** Manage blood donation camps and registrations.
9. **Notification Management:** Provide notifications related to system activities.
10. **AI Blood Assistant:** Provide information related to blood donation through an AI assistant.
11. **Feedback Management:** Allow users to submit feedback.
12. **Medical Screening and Eligibility:** Manage donor medical screening information and blood donation eligibility records.

4.2 Out of Scope

The following activities are not included in the current scope of the BloodLink system:

1. **Physical Blood Collection:** The system does not physically collect blood from donors.
2. **Laboratory Testing:** The system does not perform physical blood testing or laboratory procedures.
3. **Blood Transportation:** The system does not directly transport blood to hospitals.
4. **Medical Diagnosis and Treatment:** The system does not diagnose diseases or provide medical treatment.
5. **External Hospital System Integration:** Integration with external hospital management systems is not included unless specifically implemented.

5.Key Stakeholders

NO	Stakeholders	Responsibility
1.	Project Developer	Responsible for planning, designing, developing, and testing the BloodLink system.
2.	Project Supervisor / Lecturer	Provides academic guidance and feedback during project development.
3.	System Administrator	Manages user accounts and system operations.
4.	Blood Donors	Use the system to manage the donor information and donation-related activities.
5.	Hospital	Use the system to submit and manage blood requests.
6.	Blood Banks	Manage blood bank information and blood inventory.
7.	Donation camp organization	Manage blood donation camps and registrations.
8	System Users	Use the system's available features and provide feedback.

6. Key Deliverables

The main deliverables of the BloodLink project are:

1. **Web Application:** A completed web-based Blood Donation Management System.
2. **User Management Module**: A module to manage user registration, login, and user information.
3. **Blood Management Module**: Features to manage blood banks, blood types, and blood inventory.
4. **Blood Request Module**: Features to manage regular and emergency blood requests.
5. **Appointment Management Module**: Features to manage blood donation appointments.
6. **Donation Camp Module:** Features to manage blood donation camps and registrations.
7. **Notification Module:** A feature to provide relevant system notifications.

8. **AI Blood Assistant:** An AI-based feature that provides blood donation-related information.
9. **Project Documentation:** Documents describing the project, its features, and its design.

7. Project Constraints

The BloodLink project was developed as an individual academic project. The main project constraints were:

1. **Time Constraint:** The project had to be completed within the available academic timeframe.
2. **Resource Constraint:** The project was developed by a single developer with limited resources.
3. **Technical Constraint:** The system was developed using the selected technologies and available development tools.
4. **Budget Constraint:** The project was developed within the available budget for an academic project.
5. **Scope Constraint:** The system focused on the planned blood donation management features within the project timeframe.

8. Project Assumptions

The BloodLink project was developed based on the following assumptions:

1. **Internet Availability:** Users are expected to have access to an internet connection to use the web-based system.
2. **User Access:** Users are expected to have access to a suitable device and a compatible web browser.
3. **Accurate Information:** Users are expected to provide accurate and up-to-date information.
4. **System Availability:** The system is expected to be available when users need to access its features, subject to the hosting environment.
5. **User Knowledge:** Users are expected to have basic knowledge of using web applications.
6. **Data Availability:** Blood donors, hospitals, and blood banks are expected to provide the information required for the system's intended functions.

9. Project Success Criteria

The BloodLink project is considered successful if:

1. The main system features work according to the project requirements.
2. The system can manage blood donor information, blood requests, appointments, and blood inventory as intended.
3. The main system functions are tested, and identified major errors are addressed.
4. The system stores and manages relevant information correctly.
5. The project is completed according to the academic requirements.